

Zirui Kong

BSc in Computer Science, The Hong Kong Polytechnic University
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EDUCATION BACKGROUND

The Chinese University of Hong Kong

Master of Science in Robotics

Hong Kong, China

Sept. 2026 - Jun. 2027

The Hong Kong Polytechnic University

BSc (Hons) Computer Science; Minor in Applied Mathematics

Hong Kong, China

Sept. 2022 - Jun. 2026

PUBLICATIONS

Zirui Kong, Youqian Zhang, and Sze Yiu Chau. 2026. POSTER: Phantom Force: Injecting Adversarial Tactile Perceptions into Embodied Intelligence via EMI. In Proceedings of the ACM Asia Conference on Computer and Communications Security (ASIA CCS '26). Association for Computing Machinery, New York, NY, USA, 1922–1924. <https://doi.org/10.1145/3779208.3804894>.

RESEARCH EXPERIENCE

AI Safety Research

Research Assistant; Supervisor: Dr. ZHANG Youqian

09/2025 - Present

- ✧ Conducted research on tactile sensor security in embodied robots, focusing on EMI-based adversarial attacks on Hall-effect fingertip sensors.
- ✧ Evaluated how targeted signal injection can distort force perception and disrupt learning-based tactile classification.
- ✧ Identified robotic safety risks caused by compromised tactile sensing during manipulation tasks.

Compress-Transfer-Decompress for LLM Serving: cuSZp-Enabled CPU-GPU Data

Pipeline in vLLM

LLM Systems Engineer; Supervisor: Dr. ZHANG Zhaorui

09/2025-04/2026

- ✧ Engineered a "Compress-Transfer-Decompress" pipeline in vLLM via Monkey-Patching to overcome PCIe bottlenecks during KV cache swapping.
- ✧ Developed a high-performance C++ PyBind11 wrapper for cuSZp, implementing dynamic error-bound scaling for GPU-accelerated tensor compression.
- ✧ Achieved a $3.06 \times$ compression ratio and $2.08 \times$ Swap-In speedup across eight LLMs, safely preserving generation quality at a 10^{-4} error bound.

INTERNSHIPS

iFLYTEK Co., Ltd.

Hefei, China

Speech Feature and Internal Product Test Developer

07/2025 - 08/2025

- ✧ Designed test cases and datasets, implemented automated testing scripts and result analysis in Python, and participated in regression testing and defect localization.
- ✧ Introduced data snapshots, environment locking (containerization), and structured logging to improve reproducibility and testing efficiency.

Sina Com Technology (China) Co. LTD

Beijing, China

Agent Developer

05/2025 - 07/2025

- ✧ Designed Agent workflows and integrated tool calls, developed API wrappers and prompt templates, and conducted small-scale user testing.
- ✧ Implemented multi-layer validation, least-privilege design, and rollbackable conversation state management to mitigate tool selection errors and over-privilege risks.